FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

# Ectopic Pregnancy

Tutorial | Week 2 · Labour, delivery & the puerperium | Gynaecology | Dr Kavuma

An ectopic pregnancy implants outside the endometrial cavity, over 95% in the fallopian tube. In Uganda most women present already ruptured and shocked, so the core skill is to suspect it in every woman of reproductive age with abdominal pain or collapse, resuscitate, and get her to laparotomy fast. Stable, unruptured cases can be managed with surgery, methotrexate or careful expectant care.

**Must know before the ward round**

1. Every woman of reproductive age with abdominal pain, abnormal bleeding or collapse needs a pregnancy test. A positive test with an empty uterus means ectopic until proven otherwise.
2. Classic triad: amenorrhoea of 6–8 weeks, abdominal pain (often before the bleeding) and scanty dark vaginal bleeding. Shoulder-tip pain and fainting mean haemoperitoneum.
3. Young women keep their BP until late: tachycardia, pallor, a falling Hb and a shock index of 0.9 or more (critical at 1.4) matter more than a normal BP.
4. Ruptured and unstable: resuscitate and operate at the same time. The definitive resuscitation is clamping the bleeding tube (laparotomy, then salpingectomy). Do not wait for the BP to normalise.
5. Shocked + positive pregnancy test + peritonism = theatre. Crossmatch 4 units; do not send her for a formal scan or do a vigorous bimanual.
6. Discriminatory zone: once serum hCG is above about 1,500 IU/L (up to 3,500 by some authorities), a transvaginal scan should see an intrauterine sac (about 6,500 IU/L for a transabdominal scan).
7. PUL hCG at 0 and 48 h: rise over 63% suggests an IUP; fall over 50% suggests a failing pregnancy; anything in between needs senior review and TVUS within 24 h.
8. Methotrexate 50 mg/m² IM single dose (NICE first-line): stable, no significant pain, unruptured, mass under 35 mm, no heartbeat, hCG under 1,500 IU/L (can be offered up to 5,000), no intrauterine pregnancy, able to follow up. Check hCG on days 4 and 7 and expect a fall of 15% or more.
9. Surgery if significant pain, mass 35 mm or more, fetal heartbeat, hCG 5,000 IU/L or more, or she cannot return. In most Ugandan hospitals laparotomy and salpingectomy remain standard.
10. Salpingectomy if the other tube is healthy; salpingotomy if the other tube is damaged or absent and she wants fertility. After salpingotomy, follow hCG because trophoblast can persist.
11. Main risk factor in Uganda: tubal damage from PID (chlamydia, gonorrhoea, post-abortal and postpartum sepsis). Recurrence is about 10%, so scan early (6–7 weeks) in every future pregnancy.
12. Give anti-D to Rh-negative women having surgery; offer contraception and counselling after. No villi in an 'abortion' aspirate means think ectopic.


1 Definition
2 Epidemiology
3 Aetiology and Risk Factors
4 Pathophysiology
5 Classification / Staging
6 Clinical Features
7 Differential Diagnoses
8 Clinical Workup
9 Management
10 Complications
11 Prognosis
12 Prevention
13 Special Populations
14 Key Points
15 References

# 1. Definition

- **Ectopic pregnancy** means the fertilised ovum implants **anywhere other than the endometrial lining of the uterine cavity**. More than **95% are in the fallopian tube**.
- It is the **leading cause of maternal death in the first trimester** worldwide.
- The tube cannot stretch or form a decidua. The trophoblast erodes its wall, so the tube **ruptures and bleeds into the peritoneal cavity**, typically at **6–10 weeks**.
- **Diagnosis rests on three things:** a **pregnancy test**, **ultrasound** (empty uterus ± adnexal mass or free fluid) and **clinical suspicion**.

| Term | Meaning |
|---|---|
| **Ectopic pregnancy** | Implantation outside the endometrial cavity |
| **Ruptured vs unruptured** | Rupture = the tubal wall has burst, causing haemoperitoneum. Unruptured may still leak (tubal abortion) |
| **Tubal abortion** | The pregnancy is extruded through the fimbrial end, with bleeding from the fimbria |
| **Chronic ectopic** | Slow leak, forming an organised **pelvic haematocele** with a pelvic mass, low-grade fever, low hCG |
| **Heterotopic pregnancy** | Intrauterine + ectopic at the same time. About 1 in 30,000 spontaneous conceptions, but up to about 1 in 100 with assisted reproduction |
| **Pregnancy of unknown location (PUL)** | Positive pregnancy test but **no pregnancy seen in or outside the uterus** on TVUS. This is a *label* while you wait, not a final diagnosis |
| **Persistent trophoblast** | hCG fails to fall after conservative surgery (salpingotomy) |

> CLINICAL PEARL
> **The one thing never to forget:** a young woman who is **pale, tachycardic and has abdominal pain** has a ruptured ectopic until you have proven otherwise. **Do the pregnancy test.**

# 2. Epidemiology

- Incidence is about **1–2% of pregnancies** worldwide, higher where **sexually transmitted infections and PID** are common.
- In **African hospital series**: most cases present **ruptured** (about 71% in a Ghanaian district hospital, about 90% in Cameroon; **70–90% in hospital series** overall), often after being treated for "malaria", "PID" or "gastritis"; case fatality is **several times higher than in high-income countries**; **PID is the dominant risk factor**, with a roughly threefold increase in risk.
- **Why Ugandan women present late:**
  - early symptoms are vague;
  - women first attend drug shops, clinics or traditional healers;
  - health workers treat "malaria/UTI/PID" without a pregnancy test;
  - ultrasound and quantitative hCG are limited to HC IV and above, and **transvaginal scanning and serum hCG are rarely available outside referral and private centres**;
  - transport is slow;
  - costs (e.g. for theatre supplies or blood) delay surgery.
- **Facility realities: HC IVs can perform laparotomy**, but blood is often short. **Laparoscopy** is available only in a few referral or private hospitals. **Methotrexate** is stocked mainly at referral and cancer centres.

Repeated quantitative hCG is costly. The result: **laparotomy and salpingectomy remain the standard operation** in most Ugandan hospitals.

> **UGANDA CONTEXT**
> In Uganda, the realistic goal is **early suspicion**: a urine pregnancy test for every woman of reproductive age with abdominal pain, fainting or abnormal bleeding, at every level, including HC II and III. Then **resuscitate and refer or operate without delay**. Keep the test strips in the OPD, casualty and the gynaecology ward.

# 3. Aetiology and Risk Factors

**Normal transport.** Fertilisation occurs in the **ampulla**. **Ciliary beating and muscular peristalsis** move the embryo to the uterus over **3–4 days**, and it implants at the blastocyst stage (around day 6–7). Anything that **slows transport** (damaged cilia, adhesions, hormonal effects on motility) lets the blastocyst implant in the tube.

**Risk factors** (but half of women have none)

| Category | Examples | Why |
|---|---|---|
| **Tubal damage** | **PID** (*Chlamydia*, gonorrhoea), post-abortal or puerperal sepsis, **genital TB**, schistosomiasis | Destroys the cilia and causes intraluminal adhesions, so the embryo is transported slowly |
| **Previous ectopic** | Recurrence about 10% after one, higher after two | Same underlying tubal disease, or a scarred tube |
| **Tubal surgery** | Tubal ligation (a pregnancy after sterilisation is often ectopic), salpingotomy, reversal | Scarring or recanalisation |
| **Contraception failure** | Pregnancy with an **IUD in situ**, or with progestogen-only methods | The IUD prevents intrauterine implantation far better than tubal implantation. Progestogens slow tubal motility. The *absolute* risk is low because these methods prevent most pregnancies |
| **Infertility / ART** | IVF, ovulation induction | Tubal factor; multiple embryos |
| **Others** | Smoking, age over 35, previous pelvic or abdominal surgery (appendicitis, CS), endometriosis | Ciliary dysfunction, adhesions |

# 4. Pathophysiology

**Why the tube ruptures**

- The tube has **no decidua** (the thick, specialised endometrial layer that limits trophoblast invasion in the uterus) and only a **thin muscular wall**.
- Trophoblast invades straight into the muscularis and blood vessels.
- As the pregnancy grows, the wall thins and **ruptures**, usually:
  - **isthmus: early** (6–8 weeks), because it is narrow and non-distensible;
  - **ampulla: later** (8–12 weeks), because it is wider;
  - **interstitial (cornual): latest** (12–16 weeks), and **catastrophic**. The cornu lies where the **uterine and ovarian arteries anastomose**, and the surrounding myometrium lets the pregnancy grow larger before bursting.

**Possible outcomes** of a tubal pregnancy

1. **Spontaneous resolution** (tubal miscarriage with reabsorption). This is why expectant management works for small, declining ones.
2. **Tubal abortion** through the fimbria, with bleeding.
3. **Rupture**, with haemoperitoneum.
4. **Chronic ectopic / pelvic haematocele.**

**hCG kinetics** (why serial hCG helps)

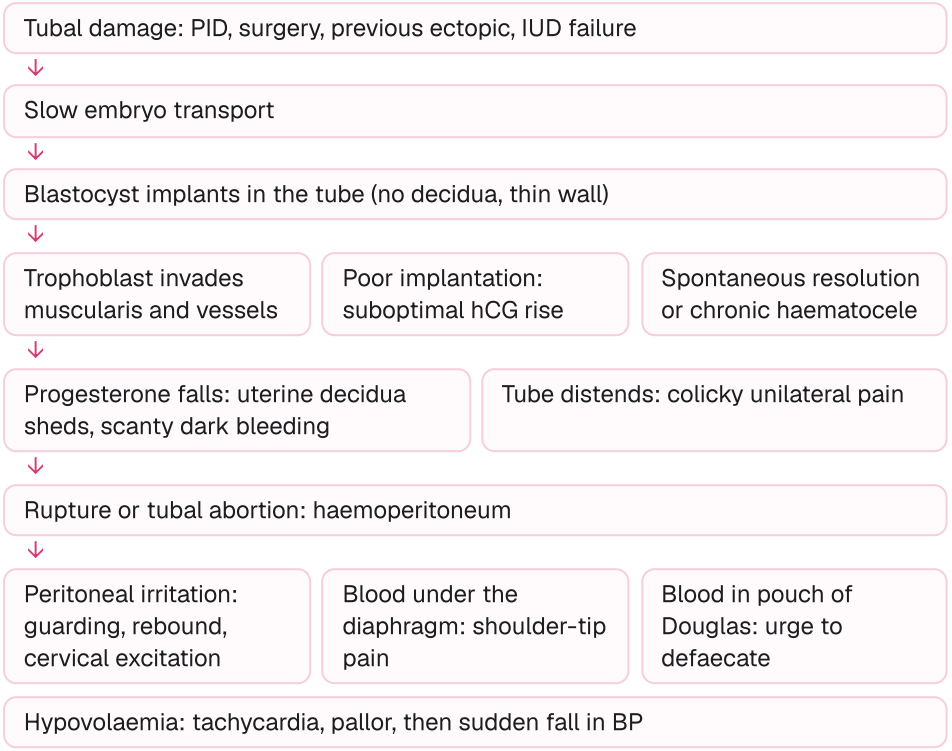
- In a viable intrauterine pregnancy, hCG roughly **doubles every 48 hours** in early pregnancy. NICE uses a **rise of more than 63% in 48 hours** as suggesting a developing IUP; ACOG uses a minimum rise of about 35%.
- A **fall of more than 50% in 48 hours** suggests a failing pregnancy.
- **Ectopic pregnancies usually show a suboptimal rise or a plateau**, because the implantation site is poor.

**Discriminatory zone.** The hCG level above which a normal IUP **should be visible** on scan:

- **TVUS:** about **1,500–2,000 IU/L** (ACOG cautions it may be as high as **3,500 IU/L**).
- **Transabdominal scan:** about **6,500 IU/L**.
- An empty uterus with hCG above the discriminatory zone strongly suggests ectopic (or a recent complete miscarriage).

**How methotrexate works.** It is a **folic acid antagonist** that inhibits **dihydrofolate reductase**, blocking DNA synthesis in rapidly dividing **trophoblast**. That explains:

- why it works best when **hCG is low** and there is **no cardiac activity** (less active, smaller trophoblast mass);
- why it is **contraindicated with an intrauterine pregnancy** and in breastfeeding;
- why its side-effects are in rapidly dividing tissues: **stomatitis, GI upset, marrow suppression, alopecia (rare)**, and **teratogenicity** (hence contraception for 3 months).




**Feature links (explaining the symptoms)**

- hCG maintains the corpus luteum → **amenorrhoea** (6–8 weeks typical).
- Poor trophoblast → less hCG → **progesterone falls** → uterine decidua sheds → **dark, scanty, "prune-juice" bleeding**; rarely a **decidual cast** mistaken for products of conception.
- Ectopic bleeding mistaken for a period → **no clear amenorrhoea** in about a third.

- Tubal distension → **colicky, unilateral pain** (visceral), typically **before** the bleeding.
- Blood in the peritoneum → **peritoneal irritation** → guarding, rebound, cervical motion tenderness.
- Blood under the diaphragm → irritates the **phrenic nerve (C3–5)** → pain referred to the C4 dermatome → **shoulder-tip pain**.
- Blood pooling in the **pouch of Douglas** → **painful defaecation / urge to defaecate**, boggy fullness on bimanual.
- Large haemoperitoneum → **distension, shifting dullness**, rarely **Cullen's sign**; free fluid in Morison's pouch on FAST.
- Haemoperitoneum → hypovolaemia → young women **compensate** with tachycardia and vasoconstriction until **30–40% of blood volume** is lost, then **BP falls suddenly**.
- Peritoneal blood → **vagally mediated bradycardia**, so a "normal" pulse does not exclude rupture.
- Ongoing bleeding → **falling Hb** on serial checks.
- Defibrinated peritoneal blood → **non-clotting blood** on culdocentesis.

# 5. Classification / Staging

### By site

| Site | Share / notes |
|---|---|
| **Tubal** (over 95%) | **Ampulla ~70–80%**, isthmus ~12%, fimbria ~11%, **interstitial (cornual) 2–3%** |
| **Non-tubal** | **Ovarian** (~3%), **cervical** (under 1%), **caesarean scar** (rising with CS rates), **abdominal** (~1%), intramural |
| **Heterotopic** | IUP + ectopic |

**By state**: ruptured (haemoperitoneum, unstable or stable) versus unruptured; tubal abortion; chronic ectopic.

**By location certainty**: PUL (no pregnancy seen) → IUP, failing PUL, or ectopic on follow-up.

**By haemodynamics**: shock index (pulse ÷ systolic BP) **0.9 or more abnormal; 1.4 or more critical**.

**Management grouping by hCG** (NICE NG126): under 1,500 IU/L → methotrexate first-line if all criteria met; 1,500–5,000 IU/L → choice of methotrexate or surgery; 5,000 IU/L or more → surgery.

# 6. Clinical Features

### History

| Ask | Why (what changes the plan) |
|---|---|
| **LMP and regularity; was the last period normal, "missed" or "unusual"?** | 6–8 weeks of amenorrhoea is typical, but about a third have no clear amenorrhoea (ectopic bleeding mistaken for a period) |
| **Pain: site, onset, character; pain or bleeding first?** | **Unilateral pain before scanty bleeding** = ectopic; bleeding first, then central cramps = miscarriage |
| **Shoulder-tip pain, fainting, dizziness, urge to defaecate** | Haemoperitoneum: urgent theatre |
| **Bleeding: amount, colour, tissue passed** | Scanty dark "prune-juice" bleeding suggests ectopic; heavy bleeding with tissue suggests miscarriage; a decidual cast can mimic POC |
| **Previous ectopic, PID/STI, tubal ligation, IUD in situ, infertility treatment, pelvic or abdominal surgery** | Raise pre-test probability; pregnancy with an IUD or after ligation is ectopic until proven otherwise |

| Ask | Why (what changes the plan) |
|---|---|
| **Contraception** | Pregnancy with an IUD or after ligation makes an ectopic much more likely |
| **Previous CS** | Caesarean scar pregnancy |
| **Any earlier scan showing an intrauterine pregnancy?** | A confirmed IUP makes ectopic very unlikely (except heterotopic after ovulation induction) |
| **Fever, dysuria, diarrhoea, vomiting, migratory pain** | Alternatives: PID, UTI, gastroenteritis, appendicitis; but always the pregnancy test first |
| **Blood group (Rh), HIV status, allergies, last meal** | Anti-D; anaesthesia and surgery planning |
| **Fertility wishes, number of living children, state of the other tube if known** | Salpingectomy vs salpingotomy; counselling |
| **Distance from the hospital, phone, ability to return** | Decides whether methotrexate or expectant care is even an option |

## Examination

Follow the order of Hutchison's Clinical Methods (24th edition). In a shocked woman, resuscitation and the pregnancy test come first (section 9), and the pelvic examination is omitted.

### Introduction, consent and preparation

- Introduce yourself, confirm identity, explain, gain consent; **chaperone** for the speculum and bimanual examination.
- **Hand hygiene**.
- **Position and exposure**: supine (flat, legs raised if shocked) for the abdomen; lithotomy or dorsal for the speculum; keep her covered.

### General inspection and vital signs

| Sign | Mechanism | What it means |
|---|---|---|
| **Pallor** | Blood loss | Haemoperitoneum or chronic anaemia |
| Cold clammy peripheries, capillary refill >2 s | Vasoconstriction | Compensated shock |
| Restlessness, confusion, drowsiness, air hunger | Cerebral hypoperfusion | Decompensating shock |
| Distended abdomen visible from the bed | Large haemoperitoneum | Theatre now |

**Vital signs**: **pulse** (tachycardia, or occasionally bradycardia), **BP** (including a **postural drop** if she is not shocked), capillary refill, respiratory rate, $SpO_2$, temperature, level of consciousness, urine output, **shock index** (0.9 or more abnormal; 1.4 or more critical).

### Hands, face and neck

Cold hands, slow capillary refill; conjunctival pallor; dry mucous membranes.

### Abdomen

- **Inspection**: distension; rarely **Cullen's sign** (periumbilical bruising); scars (previous CS, appendicectomy).
- **Palpation**: lower abdominal tenderness, usually unilateral; **guarding and rebound**.
- **Percussion**: **shifting dullness** (large haemoperitoneum).
- **Auscultation**: bowel sounds (ileus with peritonism).

### Pelvic examination

- **Speculum**: small amount of dark blood, **os closed** (in contrast to inevitable or incomplete abortion, where the os is open with tissue); decidual tissue occasionally. A **ballooned, barrel-shaped cervix** with heavy

bleeding suggests a **cervical ectopic** (do not attempt evacuation).

- **Bimanual** (gently, **once**, and **only if not in shock**, because it can rupture a friable tube):
  - **cervical excitation (motion) tenderness**;
  - adnexal tenderness or mass;
  - **fullness or bogginess in the pouch of Douglas**;
  - uterus slightly bulky but **smaller than dates**.

### Completing the examination

Thank her, hand hygiene, summarise.

**Document:** "pale, pulse X, BP Y, shock index Z; abdomen soft/guarded, shifting dullness present/absent; os closed, dark blood; cervical excitation present; right adnexal tenderness; uterus 6-week size."

> NOTE
> **Do not delay a shocked woman for a pelvic examination or a formal scan.** Positive pregnancy test + shock + signs of peritonism = theatre.

## Red flags

> RED FLAG
> - **Shock:** pulse over 100–110, systolic BP under 90, shock index 0.9 or more, cold peripheries, confusion, air hunger.
> - **Syncope, shoulder-tip pain, abdominal distension, shifting dullness.** These mean haemoperitoneum.
> - **Hb falling** on serial checks, or **free fluid in Morison's pouch** on FAST scan. The bleed is large.
> - **Pregnancy after tubal ligation or with an IUD**, plus pain.
> - **Pain at 12–16 weeks with a "normal" early scan report.** Think **interstitial (cornual)** ectopic.
> - **Heavy painless bleeding with a ballooned cervix.** Think **cervical ectopic** (do not "evacuate" it at the bedside, as massive haemorrhage follows).

> RED FLAG
> Positive pregnancy test **plus any of**: shock index 0.9 or more, fainting, shoulder-tip pain, abdominal distension, guarding or rebound, Hb falling, or free fluid in Morison's pouch = **ruptured ectopic until proven otherwise**. Senior and theatre now.

## Thinking like a clinician

- **Rule 1: pregnancy test for everyone.** Any woman aged 12–50 with abdominal pain, abnormal bleeding, collapse or "anaemia of unknown cause" gets a urine hCG. Modern urine tests detect about 25 IU/L, so a **negative test essentially excludes ectopic** (except in rare chronic ectopics).
- **Rule 2: an empty uterus + positive test = PUL or ectopic, never "complete abortion"**, unless an IUP was previously seen or the sac was seen being passed.
- **Rule 3: vital signs lie in the young.** A 22-year-old can keep a normal BP until she suddenly decompensates. Tachycardia, pallor and **a falling Hb** matter more. Some women are **bradycardic** from peritoneal blood.
- **Rule 4: in a shocked woman, surgery IS resuscitation.** You cannot fill a leaking bucket. Give fluids and blood **while transferring to theatre**. Aim for **permissive hypotension** (systolic about 80–90 mmHg, maintaining consciousness) until the tube is clamped, so you do not dilute clotting factors or "pop the clot".
- **"If X then think Y":**
  - **Positive test + adnexal mass + empty uterus:** ectopic.

- **Positive test + fever + bilateral adnexal tenderness:** still exclude ectopic before calling it PID.
- **"Incomplete abortion" but the evacuated tissue has no villi:** ectopic.
- **IUP on scan, but free fluid and severe pain:** consider **heterotopic** (especially after ovulation induction) or a **ruptured corpus luteum**.
- **Pregnancy 12–16 weeks with sudden collapse:** ruptured **interstitial or rudimentary-horn** pregnancy.

- **Common student mistakes:** forgetting the pregnancy test; accepting "PID" in a woman with a missed period; doing a vigorous bimanual examination; sending an unstable woman to the radiology department; forgetting anti-D and contraception counselling; presenting "BP normal" as reassurance.

# 7. Differential Diagnoses

| Differential | Distinguishing clinical feature | Distinguishing investigation |
|---|---|---|
| **Threatened / incomplete abortion** | Bleeding before pain; crampy central pain; **os open** (incomplete); uterine size nearer dates | Scan: IUP or retained products; villi in tissue |
| **Ruptured corpus luteum cyst** | Sudden pain mid-cycle or in early pregnancy; can cause haemoperitoneum | Scan: IUP present, ovarian cyst with free fluid |
| **Ovarian torsion** | Sudden severe unilateral pain, vomiting, adnexal mass | Scan: enlarged ovary ± reduced Doppler flow; laparoscopy or laparotomy |
| **PID / tubo-ovarian abscess** | Fever, purulent discharge, **bilateral** tenderness, pregnancy test **negative** | hCG negative; WBC raised; scan: complex adnexal mass |
| **Acute appendicitis** | Periumbilical pain migrating to the right iliac fossa, anorexia, fever; hCG negative | Clinical ± scan |
| **Urinary tract infection / ureteric colic** | Dysuria, loin pain, haematuria | Urinalysis |
| **Molar pregnancy** | Uterus large for dates, very high hCG, hyperemesis | Scan: "snowstorm" |
| **Malaria / gastroenteritis** | Fever, diarrhoea; no peritonism | mRDT; but **always do the hCG** |
| **Sickle cell crisis, ruptured spleen, perforated ulcer** | Other history and signs | As appropriate |

# 8. Clinical Workup

The two questions that drive everything are: **"Is she pregnant?"** and **"Is she bleeding into her abdomen?"** This is how the gynae emergency team at Mulago, Kawempe or Mengo would approach a woman of reproductive age with abdominal pain, abnormal bleeding or collapse. The first 5 minutes are in section 9.

### Bedside

| Test | Why / expected finding | How it changes management | Available at |
|---|---|---|---|
| **Urine hCG** | Detects about 25 IU/L. **Negative essentially excludes ectopic** (rare chronic ectopic excepted). Positive + pain = ectopic pathway | Decides the pathway in 3–5 minutes (catheter specimen if she cannot void) | All levels, including HC II |
| **HemoCue / bedside Hb** | Baseline and trend; a **falling Hb** on repeat is more telling than one value | Transfusion; ongoing bleeding → surgery | HC III and above |

| Test | Why / expected finding | How it changes management | Available at |
|---|---|---|---|
| **Bedside clotting test** | No clot by 7 minutes = coagulopathy after massive bleeding | Blood products | All levels |
| **FAST / point-of-care ultrasound** | **Free fluid in Morison's pouch** or the splenorenal space = large haemoperitoneum. Empty uterus + free fluid in a woman with a positive test is enough | Theatre | HC IV and above with an operator |
| **Culdocentesis** (needle into the pouch of Douglas through the posterior fornix) **or abdominal paracentesis** (where no ultrasound) | **Non-clotting blood** = haemoperitoneum (clotted blood in the peritoneum lyses). A **negative tap does not exclude ectopic** | Supports haemoperitoneum → theatre | All levels with a trained clinician |
| **Urine dipstick, mRDT** | Only to look for co-existing problems | **Never instead of the hCG** | All levels |

CLINICAL PEARL

**Culdocentesis:** blood aspirated from the peritoneum **does not clot**, because it has already clotted and been defibrinated by peritoneal movement. If the aspirated blood **clots in the syringe**, you probably hit a vessel.

## Basic lab

| Test | When | Expected / abnormal finding | How it changes management | Available at |
|---|---|---|---|---|
| **Hb / FBC** | Everyone | Low or falling Hb (a fall of 1–2 g/dL or more in a "stable" woman = ongoing intra-abdominal bleeding); platelets and WBC as baseline before methotrexate | Transfusion; a falling Hb in a "stable" woman → surgery | Hb at all levels; FBC at HC IV and above |
| **Group, Rhesus, crossmatch** | Everyone; **4 units if ruptured** | Rh-negative | Blood ready for theatre; **anti-D** after surgery | HC IV and above |
| **HIV, syphilis (chlamydia/gonorrhoea where available)** | Everyone | Positive | Linkage; treat her and her partner (PID is the main cause) | All levels (RDTs) |

## Imaging

| Test | When | Expected / abnormal finding | How it changes management | Available at |
|---|---|---|---|---|
| **Transvaginal ultrasound** (TVUS) | Stable woman with positive hCG | **Empty uterus**; adnexal mass separate from the ovary (**tubal ring / "bagel"**, "blob"); sac with yolk sac or embryo ± **heartbeat**; **echogenic free fluid**; **pseudosac** (central fluid without a double decidual ring); IUP (excludes ectopic in most) | Locates the pregnancy; mass size (35 mm cut-off), heartbeat and free fluid choose surgery vs methotrexate vs expectant | TVUS mainly at RRH/NRH and private units; **transabdominal** at most HC IVs |

## Specialised / referral-level

| Test | When | Expected / abnormal finding | How it changes management | Available at |
|---|---|---|---|---|
| **Serum quantitative hCG** | PUL; considering methotrexate or expectant care; after salpingotomy | Level vs **discriminatory zone** (TVUS about 1,500–2,000, up to 3,500 IU/L; transabdominal about 6,500 IU/L); 48 h trend | Above the zone with an empty uterus = ectopic (or recent complete miscarriage); **under 1,500** allows methotrexate; **5,000 or more** favours surgery | Referral hospitals and private labs; costly |
| **Repeat hCG at 48 h** | PUL | **Rise over 63%** (IUP likely); **fall over 50%** (failing); **in between** (suspect ectopic) | Rescan in 7–14 days; urine test in 2 weeks; or senior review and TVUS within 24 h | As above |
| **Serum progesterone** | Where available | Under 20 nmol/L suggests a failing pregnancy | Supports, but does not locate | Few units |
| **LFTs, creatinine, FBC** | **Before methotrexate** | Must be normal | Abnormal = methotrexate contraindicated → surgery | Referral hospitals |
| **Histology of the tube or evacuated tissue** | After surgery, where available | Chorionic villi | Confirms the diagnosis; excludes gestational trophoblastic disease | RRH/NRH |
| **Laparoscopy** | Stable woman, diagnosis uncertain or for treatment | Ectopic seen | Diagnosis and treatment in one | Few referral or private hospitals |

## Putting it together

### Model one-line summary

> "A 24-year-old P1+0 with 7 weeks of amenorrhoea, 1 day of right iliac fossa pain followed by scanty dark bleeding, and one fainting episode; pulse 118, BP 96/60 (**shock index 1.2**), pale, with right iliac fossa guarding, **positive urine hCG**, closed os and free fluid in Morison's pouch on FAST. **Working diagnosis: ruptured right tubal ectopic pregnancy with haemoperitoneum.**"

### Problem list

1. Suspected ruptured ectopic with haemorrhagic shock.
2. Anaemia (HemoCue 7.8 g/dL).
3. Rhesus status and HIV status unknown.
4. Future fertility and contraception.

**Plan:** resuscitate, crossmatch 4 units, emergency laparotomy and salpingectomy, inspect the other tube, anti-D if Rh-negative, counselling.

### Model case presentation (post-op day 1)

> "Ms K is a 26-year-old P1+0 who presented yesterday with 8 weeks of amenorrhoea, 1 day of left iliac fossa pain, shoulder-tip pain and fainting. She was **in shock** (pulse 128, BP 78/50, Hb 6.2 g/dL) with a distended, tender abdomen and a **positive urine hCG**; FAST showed free fluid in Morison's pouch. After resuscitation with 1.5 L crystalloid and 2 units of blood, she had an **emergency laparotomy**: **ruptured left ampullary ectopic**, about 2 L haemoperitoneum, **left salpingectomy**; the right tube and both ovaries were normal.

> Today she is stable, pulse 88, BP 110/70, post-transfusion Hb 8.9 g/dL, passing urine, tolerating sips. Plan: oral iron, mobilise, anti-D (she is Rh-negative), counselling on the **10% recurrence and early scan in the next pregnancy**, contraception of choice (she prefers an implant), and discharge on day 3 with review in 2 weeks."

**Worked example 1: a stable woman with hCG above the discriminatory zone.** A 29-year-old at 6 weeks, mild left pain, pulse 84, BP 118/76. TVUS: **empty uterus**, **24 mm left adnexal mass separate from the ovary, no heartbeat**, a small amount of fluid in the pouch of Douglas. Serum hCG **2,400 IU/L**.

- hCG is **above the discriminatory zone** and the uterus is empty: this is an **ectopic**, not a PUL.
- Mass under 35 mm, no heartbeat, pain mild, hCG **between 1,500 and 5,000 IU/L**: NICE says **offer a choice of methotrexate or surgery**.
- In most Ugandan hospitals (no methotrexate, uncertain follow-up), **laparotomy or laparoscopy and salpingectomy** is the safer choice. If she lives near a referral centre with methotrexate and reliable hCG, she could have **50 mg/m² IM** with hCG on days 4 and 7.

**Worked example 2: interpreting serial hCG in a PUL.** A stable woman at 5 weeks, TVUS shows nothing in or outside the uterus. hCG at 0 h is **800 IU/L**.

| hCG at 48 h | Change | Interpretation | Action |
|---|---|---|---|
| **1,400 IU/L** | Rise of 75% (over 63%) | Likely developing IUP | Rescan in 7–14 days, or when hCG passes about 1,500–2,000 IU/L |
| **1,040 IU/L** | Rise of 30% (suboptimal) | **Suspect ectopic** | Senior review and repeat TVUS **within 24 h** |
| **820 IU/L** | Plateau | **Suspect ectopic** | As above |
| **350 IU/L** | Fall of 56% (over 50%) | Failing pregnancy (miscarriage) | Urine pregnancy test in 2 weeks; return if positive or symptoms |

Calculate the change as **(new – old) ÷ old × 100**. Every woman on a PUL pathway goes home with **written safety-net advice**: return at once with pain, shoulder-tip pain, fainting or heavy bleeding.

# 9. Management

**Choosing the pathway (positive hCG + suspected ectopic)**

1. **Haemodynamically unstable or signs of rupture? → Resuscitate + emergency laparotomy (salpingectomy).**
2. **Stable: is the ectopic seen on scan?** If **no (PUL)** → serial hCG at 48 h + repeat TVUS, safety-net advice.
3. **Ectopic seen: heartbeat, mass ≥35 mm, hCG ≥5,000, significant pain, or cannot follow up? → Surgery** (laparoscopy if available, otherwise laparotomy).
4. **hCG <1,500, pain-free, mass <35 mm, no heartbeat, reliable follow-up → methotrexate** (first-line), or **expectant** if hCG is low and falling.
5. **hCG 1,500–5,000** (other criteria met) → offer **methotrexate or surgery**.

## Immediate / emergency

**First 5 minutes**

1. **Look and feel:** pale, sweaty, cold peripheries, restless or drowsy, air hunger? A distended abdomen?
2. **Vitals now:** pulse, BP, respiratory rate, $SpO_2$, temperature, capillary refill, level of consciousness, **shock index** (0.9 or more abnormal; 1.4 or more critical). A normal BP does not reassure.

3. **Urine pregnancy test immediately** (catheter specimen if she cannot void).

**Ruptured ectopic with shock: emergency algorithm (HC IV or hospital), ABCDE**

1. **Shout for help.** Inform the theatre, the anaesthetist and the lab **at once**: "ruptured ectopic, theatre now".
2. **A/B:** oxygen by face mask at 10–15 L/min. Lay her flat and raise her legs.
3. **C: two large-bore IV cannulae (14–16 G)**. Draw blood for **Hb, grouping & crossmatch (4 units)**, and a bedside clotting test.
   - **Fluids:** warmed crystalloid (Ringer's lactate or normal saline), **500 mL–1 L bolus**, then reassess.
   - **Do not over-infuse**: **permissive hypotension**, target systolic BP about 80–90 mmHg and a conscious patient until the bleeding is controlled.
   - **Transfuse early:** crossmatched blood, or **group O Rh-negative / group-specific uncrossmatched** blood if she is in extremis.
   - **Catheterise** and measure urine output.
4. **D:** level of consciousness.
5. **E:** keep her warm.
6. **Consent** (a brief, honest explanation to her and her family, including the possibility of losing the tube). **Anti-D** for Rh-negative women after surgery.
7. **Emergency laparotomy. Do not wait to "stabilise first", and do not wait for a formal scan.**

> UGANDA CONTEXT
> **At a HC III or a HC IV without a functioning theatre:** resuscitate (2 lines, fluids), make sure a **pregnancy test** is done, **call ahead**, and **refer immediately with a health worker escort**, a referral note (vitals, fluids and drugs given, time) and blood-donor relatives. Do not wait for an ultrasound. A positive pregnancy test with shock is enough to refer. In Kampala, ambulances go to Kawempe or Mulago (and Mengo, Nsambya, Rubaga for private-not-for-profit care).

## Definitive

### Emergency laparotomy for rupture

- **Anaesthesia:** ketamine-based general anaesthesia is preferred in shock; spinal is contraindicated in hypovolaemia.
- **Incision:** lower midline (fast, good access) or Pfannenstiel if the surgeon is quick with it.
- **Find and deliver the bleeding tube; clamp across the mesosalpinx** close to the tube, **and the cornual end**. This controls the haemorrhage, and the BP usually improves at once.
- **Salpingectomy:** excise the tube, ligate the pedicles with absorbable sutures (e.g. No. 1 or 2-0 polyglactin / chromic catgut).
- **Remove clots and blood** from the peritoneum. **Inspect the other tube and both ovaries** and document their state (this matters for future fertility).
- Close the abdomen in layers.
- **Autotransfusion** (when no donor blood is available). Fresh, unclotted blood from the peritoneum, with **no bowel contamination** and a recent rupture, can be scooped, **filtered through several layers of sterile gauze** into a bag containing anticoagulant (citrate), and **reinfused**. This is a recognised lifesaving technique in low-resource settings; follow the local protocol.

### Stable, unruptured tubal ectopic: surgical management

**Indications (NICE/RCOG):** significant pain; adnexal mass of **35 mm or more**; **fetal heartbeat** visible; **hCG 5,000 IU/L or more**; unable to return for follow-up; contraindication to methotrexate; signs of rupture.

| Procedure | When | Notes |
|---|---|---|
| **Salpingectomy** (removal of the tube) | **The other tube looks healthy.** This is the default | Subsequent intrauterine pregnancy rates are similar to salpingotomy when the other tube is normal (ESEP trial), with no risk of persistent trophoblast |
| **Salpingotomy** (open the tube, remove the pregnancy, leave the tube) | **The other tube is damaged or absent** and she wants future fertility | **Up to 1 in 5** need further treatment (persistent trophoblast). Check **serum hCG at 7 days, then weekly until negative**. Slightly higher recurrence in the same tube |
| **Laparoscopy vs laparotomy** | Laparoscopy is preferred where available in stable women (less pain, faster recovery). **Laparotomy** if unstable, or if laparoscopy is unavailable (most Ugandan units) | Uganda: laparotomy remains standard |

### Medical management: methotrexate

**NICE first-line criteria (all must apply):**

1. No significant pain.
2. Unruptured tubal ectopic with an **adnexal mass under 35 mm** and **no visible heartbeat**.
3. **Serum hCG under 1,500 IU/L.**
4. **No intrauterine pregnancy** on scan.
5. **Able and willing to return for follow-up.**

If hCG is **1,500–5,000 IU/L** and the other criteria are met, offer a **choice of methotrexate or surgery**.

**Contraindications:** haemodynamic instability or rupture; **intrauterine pregnancy** (including heterotopic); **breastfeeding**; liver disease or raised LFTs; renal impairment; blood dyscrasias (anaemia, leucopenia, thrombocytopenia); active peptic ulcer; active lung disease; immunodeficiency; hypersensitivity; inability to follow up.

**Monitoring: hCG on days 4 and 7**. A fall of **15% or more between day 4 and day 7** means success. It may **rise between days 1 and 4**; that is expected. Then check **weekly until negative** (typically 4–6 weeks). Up to about 1 in 7 need a second dose; consider surgery if hCG plateaus or rises again.

**Patient advice:** expect **"separation pain" on days 2–7**. It is usually mild, but **severe pain, dizziness or shoulder-tip pain means come back now** (rupture can still occur). Avoid **alcohol, folic acid supplements, NSAIDs, sun exposure and sexual intercourse** until hCG is negative. Use **effective contraception for 3 months** after methotrexate (teratogenic).

**Side-effects:** abdominal pain, nausea and vomiting, **stomatitis**, conjunctivitis, photosensitivity, transient LFT rise; rarely, marrow suppression, alopecia, pneumonitis.

> **DRUG DOSING**
>
> **Methotrexate 50 mg/m² IM.** Body surface area (Mosteller) = √(height in cm × weight in kg ÷ 3600). For example, 160 cm and 58 kg: √(9,280 ÷ 3,600) = √2.58 ≈ **1.6 m²**, giving **80 mg IM**. Methotrexate is a cytotoxic: prepare and dispose of it with cytotoxic precautions.

### Expectant management

Offer only when she is:

- **clinically stable and pain-free**;
- has a tubal ectopic **under 35 mm with no heartbeat**;
- has a **low and falling hCG**: NICE NG126 *offers* expectant management at **1,000 IU/L or below** and says *consider* it at 1,000–1,500 IU/L; RCOG GTG 21 uses **under 1,500 IU/L and falling**;

- **can return**.

Check **hCG on days 2, 4 and 7**. If it falls by **15% or more** each time, check weekly until negative. If it does not fall, or she develops pain, **switch to methotrexate or surgery**. About two-thirds or more resolve.

> **UGANDA CONTEXT**
> Medical and expectant management need **reliable serial quantitative hCG, a scan by a skilled sonographer, a phone, and money for transport to come back within hours**. For many Ugandan women these are not guaranteed. **In most Ugandan hospitals the safest option for a confirmed ectopic is surgery**, and methotrexate is used selectively at referral centres (e.g. Mulago / Kawempe) for stable women who meet the criteria and live nearby.

### Pregnancy of unknown location (PUL)

1. **Serum hCG at 0 and 48 hours** (+ progesterone if available).
2. Interpret the trend:
   - **Rise over 63%:** likely a developing IUP. **Rescan in 7–14 days**, or sooner once hCG is above the discriminatory zone.
   - **Fall over 50%:** likely a failing pregnancy. **Urine pregnancy test in 2 weeks**; return if positive.
   - **In between (the suboptimal zone): review within 24 hours** by a senior with a repeat TVUS. High suspicion of ectopic.
3. **Safety-net advice every time:** return immediately with pain, shoulder-tip pain, fainting or heavy bleeding. Give a written number.
4. **Never do an MVA or D&C "to see"** on a woman with a PUL who may have a viable IUP.

### Non-tubal ectopics (know the principles)

| Type | Clues | Management principles (specialist) |
|---|---|---|
| **Interstitial / cornual** | Later rupture (12–16 weeks), severe haemorrhage; eccentric sac with a thin myometrial mantle (under 5 mm) | Cornual resection or wedge resection at laparotomy; methotrexate if unruptured and criteria met |
| **Cervical** | Painless heavy bleeding; **barrel/hourglass cervix**; empty uterus with a sac in the cervix | Methotrexate (systemic ± local); **avoid blind evacuation**; be ready for tamponade (Foley balloon), uterine artery embolisation or ligation, and hysterectomy |
| **Caesarean scar** | Previous CS; sac in the anterior lower segment at the scar, thin myometrium | Specialist: surgical excision or aspiration under guidance; high risk of haemorrhage and accreta if left |
| **Ovarian** | Mass in the ovary; often diagnosed at surgery | Wedge resection / cystectomy; oophorectomy if necessary |
| **Abdominal** | Fetal parts easily felt, abnormal lie, painful fetal movements, empty uterus on scan; can reach late gestation | Laparotomy at a tertiary centre with blood; **leave the placenta in situ if it is attached to bowel or major vessels** (risk of torrential bleeding); follow hCG ± methotrexate |
| **Heterotopic** | IUP on scan **plus** free fluid or an adnexal mass, often after ovulation induction | Surgery for the ectopic (salpingectomy) preserving the IUP. **No methotrexate** |

## Drug table

| Drug | Dose (adult) | Route | Frequency | Duration | Key cautions |
|---|---|---|---|---|---|
| **Crystalloid** (Ringer's lactate or normal saline), warmed | 500 mL–1 L bolus, then reassess | IV | As needed | Until the tube is clamped | Permissive hypotension (systolic 80–90 mmHg); do not over-infuse |
| **Blood** | Crossmatched; group O Rh-negative / group-specific uncrossmatched in extremis | IV | As needed | | Transfuse early; post-op if Hb <7 g/dL or symptomatic |
| **Oxygen** | 10–15 L/min | Face mask | Continuous | Until stable | |
| **Ketamine** (anaesthetic induction) | Per anaesthetist | IV | | | Preferred in shock; spinal contraindicated in hypovolaemia |
| **Methotrexate** (single-dose protocol) | **50 mg/m² body surface area** (e.g. 1.6 m² gives 80 mg) | **IM** | Once (day 1) | Second dose 50 mg/m² if hCG falls **<15% between day 4 and day 7** | Baseline FBC, LFTs, creatinine, hCG, group & Rh. Cytotoxic precautions. Contraception 3 months. Avoid NSAIDs, folate, alcohol, sun |
| **Anti-D immunoglobulin** | 250 IU | IM | Once | | Rh-negative women having surgery. NICE: not needed for medical-only management; follow local policy |
| **Paracetamol** | 1 g | Oral | 6–8-hourly as needed | | **Avoid NSAIDs** after methotrexate (they reduce methotrexate clearance, increasing toxicity) |
| **Ferrous sulphate + folic acid** | Ferrous sulphate 200 mg + folic acid | Oral | 8–12-hourly | 3 months | Post-op anaemia; **no folic acid supplements during methotrexate treatment** |
| **LMWH** (thromboprophylaxis) | Per local protocol Verify *dose not stated in source* | SC | Daily | If high risk | Early mobilisation for all |

## Referral criteria

| Level | What to do | Refer up when |
|---|---|---|
| **HC II / HC III** | Urine pregnancy test for every woman of reproductive age with pain, fainting or abnormal bleeding; two lines, fluids | **Positive test with pain or shock** → call ahead and refer immediately with a health worker escort, referral note and blood-donor relatives. Do not wait for an ultrasound |
| **HC IV / General Hospital** | Transabdominal scan, Hb, crossmatch, **laparotomy and salpingectomy** | No functioning theatre or no blood → refer; stable woman wanting methotrexate or laparoscopy → RRH/NRH |
| **RRH / NRH and private** (Mulago, Kawempe, Mengo, Nsambya, Rubaga) | TVUS, serial quantitative hCG, methotrexate, laparoscopy, non-tubal ectopics (cervical, scar, abdominal) | n/a |

### Monitoring and follow-up

- **Post-op**: vitals, urine output, Hb at 24 hours; **transfuse if Hb <7 g/dL** or symptomatic; oral iron (ferrous sulphate 200 mg 8–12-hourly + folic acid) for 3 months; thromboprophylaxis (early mobilisation; LMWH if available and high risk); analgesia; anti-D.
- **After salpingotomy**: serum hCG at 7 days, then weekly until negative.
- **After methotrexate**: hCG days 4 and 7, then weekly until negative.
- **Expectant**: hCG days 2, 4 and 7, then weekly until negative.

#### Counselling (for every woman)

1. Explain what happened. **It was not her fault.** Explain what was removed and the state of the other tube.
2. **Future fertility:** about **60–70%** of women conceive an intrauterine pregnancy within 18 months when the other tube is healthy.
3. **Recurrence** is about **10%**. **In every future pregnancy, attend early for a scan (by about 6–7 weeks)** to confirm the location.
4. **Contraception:** all methods are suitable after surgery; after methotrexate, **avoid pregnancy for 3 months**; previous ectopic is **MEC 1 for IUDs and implants**, and the POP is MEC 2.
5. **Screen and treat STIs** (and the partner); HIV test.
6. Iron for anaemia. **Anti-D** if Rh-negative and surgery was done.
7. Emotional support. Grief after pregnancy loss is real.

## 10. Complications

### Disease-related

| Complication | Notes |
|---|---|
| **Haemorrhagic shock and death** | Mainly from delayed diagnosis. Most deaths are avoidable |
| **Recurrent ectopic, subfertility** | Early scan in future pregnancies; fertility referral |
| **Rh isoimmunisation** | Anti-D for Rh-negative women after surgery |
| **Chronic ectopic / pelvic haematocele** | Pelvic mass, low-grade fever |
| **Psychological morbidity** | Anxiety, grief, depression |

### Treatment-related

| Complication | Notes |
|---|---|
| **Transfusion reactions; infections (HIV/HBV risk from blood)** | Screened blood; autotransfusion where appropriate |
| **Postoperative:** wound infection, ileus, VTE, adhesions | Standard post-op care |
| **Persistent trophoblast** | After salpingotomy (up to 1 in 5). hCG follow-up; methotrexate |
| **Methotrexate toxicity** | Stomatitis, marrow suppression, hepatotoxicity, pneumonitis. Avoid NSAIDs and folate |
| **Rupture despite methotrexate or expectant care** | Safety-net advice; return with pain or dizziness |
| **Massive haemorrhage** | Blind evacuation of a cervical ectopic; removing an abdominal placenta attached to bowel or vessels |
| **Ending a viable IUP** | MVA or D&C "to see" in a PUL |

# 11. Prognosis

- Death is mainly from **delayed diagnosis**; most deaths are avoidable with a pregnancy test and prompt surgery.
- About **60–70%** conceive an intrauterine pregnancy within 18 months when the other tube is healthy.
- **Recurrence about 10%** after one ectopic, higher after two.
- Methotrexate: up to about 1 in 7 need a second dose; expectant management: about two-thirds or more resolve.
- Salpingotomy: up to 1 in 5 need further treatment for persistent trophoblast.

# 12. Prevention

- **Primary**: prevent and treat **STIs and PID** promptly (and treat partners); safe abortion care and prevention of post-abortal and puerperal sepsis.
- **Secondary**: **pregnancy test for every woman of reproductive age** with abdominal pain, fainting or abnormal bleeding, at every level; keep test strips in OPD, casualty and the gynaecology ward.
- **Tertiary**: early scan at 6–7 weeks in every future pregnancy; effective contraception; STI screening and partner treatment after an ectopic.

# 13. Special Populations

- **IUD in situ or after tubal ligation**: any pregnancy is ectopic until proven otherwise.
- **Assisted reproduction / ovulation induction**: heterotopic pregnancy up to about 1 in 100; an IUP on scan does not exclude ectopic. No methotrexate with a coexisting IUP.
- **Breastfeeding**: methotrexate contraindicated.
- **Rh-negative**: anti-D 250 IU with surgery.
- **Previous CS**: caesarean scar pregnancy (specialist).
- **Single remaining or damaged other tube**: salpingotomy to preserve fertility, with hCG follow-up.
- **Women who cannot return** (distance, no phone, cost): surgery rather than methotrexate or expectant care.
- **HIV**: test, link to care; PID is the main cause, so screen and treat STIs.

# 14. Key Points

### High-yield points

- Pregnancy test for every woman of reproductive age with pain, bleeding or collapse.
- Positive test + empty uterus = ectopic (or PUL), never "complete abortion" without a previous IUP.
- Pain before scanty dark bleeding; shoulder-tip pain and fainting = haemoperitoneum.
- Vital signs lie in the young: pulse, pallor, shock index and a falling Hb.
- Shocked + positive test + peritonism = theatre; clamping the tube is the resuscitation.
- Discriminatory zone: TVUS about 1,500–2,000 (up to 3,500) IU/L; transabdominal about 6,500.
- PUL: rise >63% = IUP likely; fall >50% = failing; in between = suspect ectopic.
- Methotrexate 50 mg/m² IM if criteria met; hCG days 4 and 7, fall ≥15%.
- Salpingectomy if the other tube is healthy; salpingotomy if not, with hCG follow-up.
- Anti-D, contraception, STI treatment, early scan next time.

## Numbers & doses

| Item | Value |
|---|---|
| Tubal share of ectopics | Over **95%**; ampulla about 70–80% |
| Typical presentation | **6–8 weeks** of amenorrhoea |
| Urine hCG sensitivity | About **25 IU/L** |
| Shock index | **0.9 or more** abnormal; **1.4 or more** critical |
| Crossmatch for ruptured ectopic | **4 units** |
| Permissive hypotension target | Systolic about **80–90 mmHg**, conscious, until the tube is clamped |
| Discriminatory zone | TVUS about **1,500–2,000 IU/L** (up to 3,500); transabdominal about **6,500 IU/L** |
| PUL 48 h hCG | Rise **over 63%** = IUP likely; fall **over 50%** = failing; in between = suspect ectopic |
| Methotrexate first-line | hCG **under 1,500**, mass **under 35 mm**, no heartbeat, no pain, no IUP, can return |
| Methotrexate or surgery (choice) | hCG **1,500–5,000 IU/L** |
| Surgery indicated | Mass **35 mm or more**, heartbeat, hCG **5,000 or more**, significant pain, rupture |
| Methotrexate dose | **50 mg/m² IM** single dose |
| Methotrexate success | hCG fall of **15% or more** between **day 4 and day 7** |
| Avoid pregnancy after methotrexate | **3 months** |
| Expectant management | hCG **1,000 IU/L or below** (NICE offers; consider up to 1,500) and falling |
| After salpingotomy | hCG at **7 days**, then weekly; up to **1 in 5** need further treatment |
| Transfusion trigger post-op | Hb **under 7 g/dL** or symptomatic |
| Anti-D | **250 IU IM** with surgery if Rh-negative |
| Recurrence | About **10%**; early scan at **6–7 weeks** next time |

## Exam traps

- **"BP is 110/70, so she is not bleeding much."** Wrong: young women compensate until 30–40% blood loss. Look at the pulse, pallor, shock index and a **falling Hb**.
- **"Resuscitate until stable, then operate."** Wrong: in a ruptured ectopic, **clamping the tube is the resuscitation**; do both together.
- **"Fever and bilateral tenderness: treat as PID."** Wrong until the **pregnancy test** is negative.
- **"Empty uterus + positive test after bleeding = complete miscarriage."** Wrong without a previous IUP: it is a **PUL/ectopic**.
- **"The hCG is low, so it cannot be ruptured."** Wrong: ectopics can rupture at **any hCG level**.
- **"Negative culdocentesis excludes ectopic."** Wrong: only a positive (non-clotting blood) tap is useful.
- **"Give ibuprofen for separation pain after methotrexate."** Wrong: **avoid NSAIDs** (reduce methotrexate clearance); use paracetamol.
- **"Do an MVA to see if there are villi."** Wrong in a PUL: you may end a **viable IUP**.

## Questions seniors ask

### Viva questions

1. *What is the commonest site of ectopic pregnancy?* The **ampulla** of the fallopian tube.
2. *Why does shoulder-tip pain occur?* Blood irritates the diaphragm; the phrenic nerve (C3–5) refers pain to the C4 dermatome.

3. *What is the discriminatory zone?* The hCG level (about 1,500–2,000 IU/L on TVUS) above which an intrauterine sac should be visible.
4. *Give the NICE criteria for first-line methotrexate.* No significant pain; unruptured; mass under 35 mm; no heartbeat; hCG under 1,500; no IUP; able to follow up.
5. *How do you monitor after methotrexate?* hCG on days 4 and 7; a fall of 15% or more is success, then weekly until negative. A second dose if the fall is under 15%.
6. *Salpingectomy or salpingotomy?* Salpingectomy if the other tube is healthy. Salpingotomy if the other tube is damaged or absent, with hCG follow-up.
7. *A shocked woman with a positive pregnancy test: resuscitate first, then theatre?* **Both at once.** Surgery is the resuscitation. Do not delay for the BP to normalise.
8. *What does non-clotting blood on culdocentesis mean?* Haemoperitoneum (defibrinated blood).

**On the ward round**

- **What is your first investigation in a young woman with abdominal pain?** A **urine pregnancy test**, at every level of care.
- **What is the discriminatory zone and why does it matter?** The hCG (about 1,500–2,000 IU/L on TVUS) above which an intrauterine sac should be seen; an empty uterus above it strongly suggests ectopic.
- **How do you interpret a 48 h hCG?** Rise over 63% = likely IUP; fall over 50% = failing pregnancy; in between = suspect ectopic, senior review and TVUS within 24 h.
- **Who can have methotrexate?** Stable, no significant pain, unruptured, mass under 35 mm, no heartbeat, hCG under 1,500 (up to 5,000 as a choice), no IUP, normal FBC/LFTs/renal function, able to return.
- **What will you look at in theatre besides the ruptured tube?** The **other tube and both ovaries**: their state decides salpingectomy vs salpingotomy and her fertility counselling.
- **What if there is no blood in the bank?** Resuscitate with crystalloid, ask relatives to donate, and consider **autotransfusion** of fresh, uncontaminated haemoperitoneum filtered through gauze into citrate.
- **What do you tell her before discharge?** It was not her fault; about 10% recurrence; **early scan at 6–7 weeks** next pregnancy; contraception of choice; STI treatment for her and her partner; anti-D if Rh-negative.

## Mnemonics

**Risk factors: "PID SITE"**

- **P**revious ectopic
- **I**UD / progestogen-only failure
- **D**amaged tubes (PID, TB, surgery)
- **S**terilisation / tubal surgery
- **I**nfertility / IVF
- **T**obacco (smoking)
- **E**lderly (over 35) / **E**ndometriosis

**Methotrexate criteria: "No PAIN, No BEAT, Under 35 mm, Under 1,500, Can come back"**

**Methotrexate avoid-list: "SAFE-N"**: **S**un, **A**lcohol, **F**olate, **E**xpectancy (pregnancy) for 3 months, **N**SAIDs.

**Classic presentation: "A-P-B"** (Amenorrhoea, Pain, Bleeding), with Pain before Bleeding.

**Ruptured ectopic: "Clamp is the cure"**: laparotomy is resuscitation.

### OSCE tips

> EXAM FAVOURITE
> **OSCE station ("young woman with abdominal pain"):** your first investigation must be the **pregnancy test**. Examiners fail candidates who order an abdominal X-ray or "malaria slide" first. In the history, always ask about the LMP, contraception, STI/PID, previous ectopic and **shoulder-tip pain / fainting**.

- In the examination station, say you would **not** do a bimanual examination in a shocked woman, and that a bimanual, when done, is gentle and once.
- In a counselling station, cover: what happened, not her fault, the state of the other tube, 10% recurrence, early scan next time, contraception, STI treatment for her and her partner.

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SELF-TEST

# Flashcards

26 cards. Cover the right-hand column and test yourself.

| # | Question | Answer |
|---|---|---|
| 1 | Commonest site of ectopic pregnancy | **Ampulla** of the fallopian tube (~70–80%); >95% of ectopics are tubal. |
| 2 | Which tubal ectopic ruptures latest and bleeds most catastrophically, and why? | **Interstitial (cornual)** – ruptures at 12–16 weeks because surrounding myometrium allows growth; sits at the **uterine–ovarian artery anastomosis**. |
| 3 | Classic triad of ectopic pregnancy | **Amenorrhoea** (6–8 wk), **abdominal pain** (often unilateral, preceding bleeding), **scanty dark vaginal bleeding**. |
| 4 | Why does shoulder-tip pain occur in ruptured ectopic? | Blood irritates the diaphragm → **phrenic nerve (C3–5)** → pain referred to the C4 dermatome (shoulder). |
| 5 | Why does vaginal bleeding occur in ectopic pregnancy? | Low hCG production → **falling progesterone** → shedding of the uterine **decidua** (sometimes as a decidual cast). |
| 6 | Most important risk factor for ectopic pregnancy in Uganda | **Tubal damage from PID** (chlamydia, gonorrhoea; also post-abortal/puerperal sepsis, genital TB). |
| 7 | First investigation in any woman of reproductive age with abdominal pain or collapse | **Urine pregnancy test (hCG)** – at every level of care. |
| 8 | Discriminatory zone for TVUS | hCG ~**1,500–2,000 IU/L** (ACOG: up to 3,500) above which an intrauterine sac should be seen; ~6,500 IU/L for transabdominal scan. |
| 9 | hCG 48-hour trends in PUL (NICE) | Rise **>63%** → likely IUP; fall **>50%** → failing pregnancy; in between → **suspect ectopic**, senior review within 24 h. |
| 10 | Ultrasound features of tubal ectopic | **Empty uterus**, adnexal mass separate from ovary (**bagel/tubal ring** or 'blob' sign) ± yolk sac/embryo/heartbeat, **echogenic free fluid**; beware pseudosac. |
| 11 | NICE criteria for first-line methotrexate | No significant pain; unruptured; **mass <35 mm; no heartbeat; hCG <1,500 IU/L**; **no IUP**; able to return for follow-up. (hCG 1,500–5,000: offer choice of MTX or surgery.) |
| 12 | Methotrexate dose for ectopic pregnancy | **50 mg/m² IM** single dose (e.g. 1.6 m² → 80 mg). |
| 13 | Monitoring after methotrexate | hCG **days 4 and 7**: fall **≥15%** between day 4 and 7 = success → weekly until negative. <15% → second dose or surgery. |
| 14 | What to avoid after methotrexate | **Sun, alcohol, folic acid, NSAIDs, sexual intercourse** until resolved; **pregnancy for 3 months**. |
| 15 | Contraindications to methotrexate (name 6) | Haemodynamic instability/rupture, **intrauterine pregnancy**, **breastfeeding**, liver/renal disease, blood dyscrasia, peptic ulcer, active lung disease, immunodeficiency, unable to follow up. |
| 16 | Salpingectomy vs salpingotomy – when each? | **Salpingectomy** if contralateral tube healthy (default). **Salpingotomy** if other tube damaged/absent and fertility desired – follow hCG (up to 1 in 5 need further treatment). |
| 17 | Management principle of ruptured ectopic in shock | **Resuscitate and operate simultaneously** – laparotomy and clamping the bleeding tube IS the resuscitation; permissive hypotension until clamp. |
| 18 | What is autotransfusion in ruptured ectopic? | Reinfusion of fresh, uncontaminated **haemoperitoneum blood filtered through gauze into an anticoagulant bag** – lifesaving when donor blood is unavailable. |

| # | Question | Answer |
|---|---|---|
| 19 | Why does blood from culdocentesis not clot? | Blood in the peritoneum has already clotted and been **defibrinated** – non-clotting blood indicates haemoperitoneum. |
| 20 | Recurrence risk after one ectopic and key advice | About **10%**; attend for an **early scan (~6–7 weeks)** in every future pregnancy. |
| 21 | Expectant management criteria for tubal ectopic | Stable, **pain-free**, mass **<35 mm no heartbeat**, **hCG low and falling** (NICE offers it at ≤1,000 IU/L, consider 1,000–1,500; RCOG <1,500), can return; hCG days 2, 4, 7 (≥15% fall each). |
| 22 | PUL: hCG 800 IU/L at 0 h and 1,040 IU/L at 48 h. Interpret and act. | Rise of **30%** (below 63%): **suboptimal**, so **suspect ectopic**. Senior review and repeat **TVUS within 24 h**; safety-net advice. |
| 23 | Stable woman: empty uterus on TVUS, 24 mm adnexal mass, no heartbeat, hCG 2,400 IU/L. Options (NICE)? | hCG is above the discriminatory zone, so this is an **ectopic**. With hCG **1,500–5,000 IU/L**, offer a **choice of methotrexate 50 mg/m² IM or surgery**. In most Ugandan hospitals, surgery (salpingectomy) is the safer default. |
| 24 | Positive pregnancy test, pulse 124, BP 90/60, free fluid in Morison's pouch. Next steps? | **Ruptured ectopic.** Call theatre and anaesthetist, $O_2$, two 14–16 G cannulae, **crossmatch 4 units**, warmed crystalloid 500 mL–1 L, permissive hypotension (SBP 80–90), catheter, consent and **emergency laparotomy + salpingectomy**. No formal scan, no vigorous bimanual. |
| 25 | Tests required before giving methotrexate for ectopic | **FBC, LFTs, creatinine**, baseline **serum hCG**, **group and Rhesus**, and a TVUS confirming **no intrauterine pregnancy**. Abnormal results mean surgery instead. |
| 26 | Is a normal BP reassuring in a young woman with suspected ectopic? | **No.** Young women compensate until about **30–40% blood loss**, then BP falls suddenly. Use pulse, pallor, a **shock index of 0.9 or more** (critical 1.4) and a **falling Hb**; some are bradycardic from peritoneal blood. |

SELF-TEST

# Single best answer questions

15 questions. The answers and explanations follow at the end.

**1.** A 24-year-old is brought to the casualty of a regional referral hospital after collapsing at home. She has lower abdominal pain and 7 weeks of amenorrhoea. Pulse 132, BP 70/40, cold peripheries, distended tender abdomen. Urine hCG positive. What is the MOST appropriate next step?

A Formal pelvic ultrasound in the radiology department
B Intramuscular methotrexate 50 mg/m²
C Serum quantitative hCG and repeat in 48 hours
D Resuscitate with IV fluids and blood while arranging immediate laparotomy
E Resuscitate until systolic BP is above 110 mmHg, then consider surgery

**2.** A 30-year-old at 6 weeks by LMP has mild left iliac fossa pain. She is stable. TVUS: empty uterus, 22 mm left adnexal mass with no cardiac activity, no free fluid. Serum hCG 1,100 IU/L. FBC, LFTs and creatinine are normal. She lives 3 km from the hospital and has a phone. What is the MOST appropriate management according to NICE?

A Emergency laparotomy and salpingectomy
B Methotrexate 50 mg/m² IM with hCG on days 4 and 7
C MVA to exclude an intrauterine pregnancy
D Discharge with no follow-up
E Salpingotomy by laparotomy

**3.** A woman given methotrexate for a tubal ectopic has hCG 1,300 IU/L on day 1, 1,450 IU/L on day 4 and 1,320 IU/L on day 7. She is pain-free and stable. What is the MOST appropriate next step? Advanced

A Reassure – treatment has succeeded
B Start folic acid to reduce side effects
C Immediate laparotomy
D Repeat hCG in 4 weeks
E Give a second dose of methotrexate 50 mg/m² (or offer surgery)

**4.** A 28-year-old at 7 weeks presents with vaginal spotting and lower abdominal pain. Urine hCG positive. She is stable. TVUS: empty uterus, no adnexal mass, no free fluid. Serum hCG 900 IU/L; 48 hours later 1,020 IU/L. What is the MOST likely interpretation?

A Normal intrauterine pregnancy – rescan in 2 weeks
B Complete miscarriage – discharge
C Suboptimal rise – high suspicion of ectopic; senior review and repeat TVUS within 24 hours
D Molar pregnancy
E Perform MVA to diagnose

**5.** At laparotomy for a ruptured right tubal ectopic in a 27-year-old P0 who wants children, the left tube appears normal. What is the MOST appropriate procedure?

A Right salpingectomy
B Right salpingotomy to preserve fertility
C Bilateral salpingectomy
D Total abdominal hysterectomy
E Right salpingo-oophorectomy

**6.** A 32-year-old with a previous right salpingectomy for ectopic now has a left tubal ectopic, unruptured, 30 mm, hCG 6,200 IU/L, with a visible heartbeat. She is stable and strongly desires future pregnancy. Laparoscopy is available. What is the BEST option? Advanced

- A Methotrexate 50 mg/m² IM
- B Expectant management
- C Laparoscopic left salpingotomy with hCG follow-up
- D Laparoscopic left salpingectomy
- E MVA

**7.** A 23-year-old presents to a HC III with lower abdominal pain, fever 37.9 °C and vaginal discharge. The clinician diagnoses PID and prescribes ceftriaxone, doxycycline and metronidazole. Her LMP was 7 weeks ago. What is the MOST important omission?

- A HIV test
- B Urinalysis
- C Malaria RDT
- D Urine pregnancy test
- E High vaginal swab

**8.** Which patient is MOST suitable for expectant management of a tubal ectopic?

- A Stable, pain-free, 20 mm mass without heartbeat, hCG 800 falling to 520 over 48 h, lives nearby
- B Stable, 38 mm mass, hCG 1,200 rising
- C Mild pain, 25 mm mass with heartbeat, hCG 2,500
- D Shoulder-tip pain, free fluid in Morison's pouch
- E Pain-free, 15 mm mass, hCG 3,800, lives 80 km away

**9.** A 26-year-old on methotrexate day 3 for ectopic presents with moderate lower abdominal pain. Pulse 84, BP 118/76, abdomen soft with mild tenderness, no guarding. Hb unchanged from baseline. TVUS: no increase in free fluid. What is the MOST appropriate management? Advanced

- A Immediate laparotomy
- B Discharge with no follow-up
- C Ibuprofen 400 mg 8-hourly
- D Give a second dose of methotrexate
- E Paracetamol, observation and reassessment – likely separation pain

**10.** Which is an absolute contraindication to methotrexate in a woman with a tubal ectopic?

- A hCG 1,200 IU/L
- B Breastfeeding a 6-month-old baby
- C Adnexal mass of 25 mm
- D Nulliparity
- E Previous caesarean section

**11.** A 29-year-old P2 had an emergency laparotomy for ruptured ectopic at a HC IV where no blood is available. There is 1.5 L of fresh, dark, unclotted blood in the peritoneum, no bowel injury, and she remains hypotensive. What is the MOST appropriate option to improve her circulation? Advanced

- A Autotransfusion: collect the blood, filter it through gauze into an anticoagulant bag and reinfuse
- B Suction the blood to waste and give more saline
- C Give adrenaline infusion
- D Delay closing and wait for donor blood from the regional hospital
- E Give tranexamic acid only

**12.** A 35-year-old G3P2 with two previous caesarean sections is 7 weeks pregnant. TVUS shows an empty uterine cavity and cervical canal, with a gestational sac embedded in the anterior lower uterine segment at the scar, thin overlying myometrium. What is the diagnosis and MOST appropriate action? Advanced

- A Normal low-lying intrauterine pregnancy – routine ANC
- B Cervical ectopic – immediate D&C
- C Caesarean scar ectopic – refer to a specialist centre; avoid blind evacuation
- D Incomplete abortion – MVA at the HC IV
- E Missed abortion – misoprostol 800 µg

**13.** A woman has a salpingotomy for tubal ectopic. What follow-up is MOST appropriate?

- A No follow-up is needed
- B Methotrexate prophylaxis routinely
- C Ultrasound at 6 weeks only
- D Hysterosalpingogram at 2 weeks
- E Serum hCG at 7 days, then weekly until negative

**14.** A 22-year-old P0 who had ovulation induction with clomiphene has a live intrauterine pregnancy at 7 weeks on scan, but presents with severe right iliac fossa pain, pulse 118 and free fluid in the pouch of Douglas. What is the MOST likely diagnosis? Advanced

- A Threatened abortion
- B Round ligament pain
- C Acute appendicitis
- D Heterotopic pregnancy
- E Ovarian hyperstimulation – no action needed

**15.** A 20-year-old presents to an HC III with 8 weeks' amenorrhoea, abdominal pain and fainting. Urine hCG is positive, pulse 124, BP 86/54. The HC III has no theatre or ultrasound; the HC IV is 40 minutes away. What is the BEST action?

- A Wait for ultrasound confirmation before referral
- B Two IV lines, fluids, and urgent escorted referral with a note and blood-donor relatives, calling ahead
- C Give oral analgesics and observe overnight
- D Treat for malaria and review in the morning
- E Give misoprostol for suspected incomplete abortion

## Answers & explanations

> **1. D** · Resuscitate with IV fluids and blood while arranging immediate laparotomy
>
> Positive hCG + shock + peritonism = **ruptured ectopic → resuscitate and operate simultaneously**. **A/C**: delaying for imaging/serial tests risks death; bedside FAST at most. **B**: MTX is contraindicated in rupture/instability. **E**: waiting for normal BP is futile – she keeps bleeding; clamping the tube is the definitive resuscitation.

> **2. B** · Methotrexate 50 mg/m² IM with hCG on days 4 and 7
>
> She meets **NICE first-line MTX criteria**: no significant pain, unruptured, mass <35 mm, no heartbeat, hCG <1,500, no IUP, able to follow up. **A/E**: surgery is not required (though acceptable if she prefers or MTX unavailable). **C**: MVA could disrupt a possible very early IUP and doesn't treat the ectopic. **D**: follow-up is essential – ectopics can rupture.

**3. E** · Give a second dose of methotrexate 50 mg/m² (or offer surgery)

Day 4 → day 7 fall = (1,450 – 1,320)/1,450 ≈ **9% (<15%) → inadequate response: second dose or surgery**. **A**: success requires ≥15% fall. A rise between days 1 and 4 is expected. **C**: she is stable, no rupture. **D**: too long an interval. **B**: folic acid antagonises methotrexate.

**4. C** · Suboptimal rise – high suspicion of ectopic; senior review and repeat TVUS within 24 hours

A **13% rise** is in the suboptimal zone (neither >63% rise nor >50% fall) → **suspect ectopic**. **A**: viable IUP rises >63%. **B**: a complete miscarriage shows falling hCG. **D**: mole gives very high hCG and a snowstorm uterus. **E**: MVA may disrupt a viable IUP and misses the ectopic.

**5. A** · Right salpingectomy

With a healthy contralateral tube, **salpingectomy** gives similar future IUP rates to salpingotomy without persistent trophoblast; a ruptured tube is usually not salvageable. **B**: salpingotomy is for a damaged/absent contralateral tube. **C/D**: unnecessary and destroy fertility. **E**: the ovary should be preserved unless involved.

**6. C** · Laparoscopic left salpingotomy with hCG follow-up

Heartbeat and hCG >5,000 → **surgery**, and with the **only remaining tube**, **salpingotomy** preserves the chance of natural conception; follow hCG weekly for persistent trophoblast. **A/B**: heartbeat and high hCG make MTX/expectant inappropriate. **D**: would leave her needing IVF. **E**: not treatment for ectopic.

**7. D** · Urine pregnancy test

Any woman of reproductive age with pelvic pain and a missed period needs a **pregnancy test** – ectopic pregnancy is commonly misdiagnosed as PID in Uganda. **A/B/C/E** are reasonable but none would detect the potentially fatal diagnosis. Note doxycycline is also contraindicated in pregnancy.

**8. A** · Stable, pain-free, 20 mm mass without heartbeat, hCG 800 falling to 520 over 48 h, lives nearby

Expectant criteria: **stable, pain-free, mass <35 mm without heartbeat, hCG low and falling (NICE ≤1,000 IU/L; consider up to 1,500), reliable follow-up** – option A. **B**: mass ≥35 mm and rising hCG. **C**: heartbeat. **D**: haemoperitoneum → surgery. **E**: hCG too high and follow-up unreliable.

**9. E** · Paracetamol, observation and reassessment – likely separation pain

Pain on days 2–7 after MTX is common (**separation pain**). With stable vitals, stable Hb and no new free fluid: **paracetamol and observation**, low threshold for surgery if it worsens. **A**: no evidence of rupture. **C**: NSAIDs increase methotrexate toxicity. **D**: second dose is decided by the day 4–7 hCG fall. **B**: she still needs hCG monitoring.

**10. B** · Breastfeeding a 6-month-old baby

**Breastfeeding** is a contraindication (MTX is excreted in milk). **A/C** are within criteria. **D/E** are irrelevant to MTX suitability.

**11. A** · Autotransfusion: collect the blood, filter it through gauze into an anticoagulant bag and reinfuse

**Autotransfusion** of fresh, uncontaminated haemoperitoneum is a recognised lifesaving technique where donor blood is unavailable. **B**: excess crystalloid causes dilutional coagulopathy without oxygen-carrying capacity. **C**: vasopressors don't replace volume. **D**: dangerous delay. **E**: TXA does not replace lost red cells.

**12. C** · Caesarean scar ectopic – refer to a specialist centre; avoid blind evacuation

Sac at the CS scar with thin myometrium = **caesarean scar pregnancy** – high risk of haemorrhage, rupture and placenta accreta → **specialist management**. **A**: missing it risks catastrophic haemorrhage later. **B**: the sac is in the scar, not the cervix; blind D&C causes massive bleeding. **D/E**: not an abortion; blind evacuation/misoprostol is dangerous.

**13. E** · Serum hCG at 7 days, then weekly until negative

After salpingotomy, **persistent trophoblast** occurs in up to 1 in 5 – check **hCG at 7 days then weekly until negative** (NICE). **A**: misses persistent trophoblast. **C**: ultrasound doesn't detect persistent trophoblast reliably. **D**: HSG is for later tubal patency assessment, not early follow-up. **B**: not routine.

**14. D** · Heterotopic pregnancy

IUP + haemoperitoneum after ovulation induction → **heterotopic pregnancy** (risk rises with fertility treatment). Manage surgically (salpingectomy) preserving the IUP. **A**: does not cause haemoperitoneum. **C**: possible but free fluid + shock points to bleeding. **B**: benign, no shock. **E**: OHSS causes ascites but tachycardia with free fluid requires urgent evaluation, and after clomiphene OHSS is uncommon.

**15. B** · Two IV lines, fluids, and urgent escorted referral with a note and blood-donor relatives, calling ahead

Suspected ruptured ectopic: **resuscitate (2 lines, fluids), call ahead and refer urgently with escort, note and donors**. **A/C/D** are fatal delays. **E**: misoprostol does nothing for an ectopic and presumes the wrong diagnosis.

SELF-TEST

# Clinical cases

Work through each step before reading the model answer.

## Case 1: Collapse in casualty: ruptured ectopic

A 26-year-old P1+0 is carried into Mengo Hospital casualty by her husband after fainting at the market. She complains of lower abdominal pain since morning and pain in her right shoulder. She is pale and sweaty.

**1. What are your immediate actions in the first 5 minutes?**

> Call for help; **ABC**; oxygen; lie flat, legs up; **2 large-bore IV cannulae**; bloods for **Hb, group & crossmatch 4 units**, bedside clotting; **urine hCG** (catheter specimen if needed); warmed crystalloid bolus; vitals every 5–15 min; alert theatre, anaesthetist, lab.

**2. What focused history do you obtain (from her or her husband)?**

> **LMP** (7–8 weeks ago?), pain onset and site (unilateral before bleeding?), **shoulder-tip pain, syncope**, vaginal bleeding, contraception (IUD? ligation?), previous ectopic/PID/STI, infertility treatment, previous surgery, Rh group, allergies, last meal.

**3. Pulse 128, BP 76/48, distended abdomen with generalised tenderness and shifting dullness; urine hCG positive; HemoCue Hb 6.0 g/dL; bedside FAST shows free fluid in Morison's pouch. Diagnosis and differentials?**

> **Ruptured ectopic pregnancy with haemorrhagic shock.** Differentials: ruptured corpus luteum cyst with IUP, ruptured spleen, perforated viscus – but hCG + haemoperitoneum makes ectopic by far the most likely.

**4. Outline definitive management.**

> **Resuscitate and operate simultaneously**: permissive hypotension (SBP ~80–90), transfuse (O-neg/uncrossmatched if needed), consent, **ketamine GA**, **emergency laparotomy**: clamp the bleeding tube, **salpingectomy**, remove blood, inspect other tube/ovaries; autotransfusion if no blood; catheter, fluid chart.

**5. What post-operative care and counselling does she need?**

> Vitals, urine output, **Hb at 24 h** (transfuse if <7 g/dL), **oral iron + folate**, analgesia, early mobilisation/VTE prevention, **anti-D if Rh-negative**. Counsel: what happened, **10% recurrence, early scan in next pregnancy**, fertility outlook, **contraception**, STI screening and partner treatment, emotional support; review in 2 weeks.

**Take-away:** Positive pregnancy test + shock = ruptured ectopic; the clamp in theatre is the resuscitation.

## Case 2: Stable ectopic: to operate or not?

A 31-year-old G2P1, a teacher living 2 km from Kawempe National Referral Hospital, is referred with 6 weeks of amenorrhoea and mild intermittent right iliac fossa pain. She is well, pulse 78, BP 122/78. Urine hCG positive.

**1. What investigations do you request?**

> **TVUS** (uterus, adnexa, free fluid), **serum quantitative hCG** (repeat at 48 h if PUL), Hb/FBC, **group & Rh**, and — if medical management is considered — **LFTs and creatinine**.

**2. TVUS: empty uterus, 24 mm right adnexal mass separate from the ovary with a yolk sac but no heartbeat, minimal free fluid. hCG 1,350 IU/L. Bloods normal. What are her options?**

> Meets NICE criteria for **methotrexate (first-line)**: stable, no significant pain, mass <35 mm, no heartbeat, hCG <1,500, no IUP, can follow up. Alternatives: **surgery (salpingectomy)**, or expectant if hCG falling. Discuss pros/cons: MTX avoids surgery but requires 4–6 weeks of follow-up and carries a small rupture risk.

**3. She chooses methotrexate. Prescribe it and give her the advice.**

> **Methotrexate 50 mg/m² IM** (e.g. BSA 1.7 m² → 85 mg), cytotoxic precautions. Advice: **paracetamol not NSAIDs**, avoid **alcohol, folic acid, sunlight, intercourse**; **contraception for 3 months**; expect some pain days 2–7; **return immediately** with severe pain, dizziness, shoulder-tip pain; hCG on **days 4 and 7**.

**4. Day 4 hCG 1,480; day 7 hCG 1,150 IU/L. Interpretation and plan?**

> Fall from day 4 to 7 = (1,480 – 1,150)/1,480 ≈ **22% (≥15%) → responding**. The day-4 rise is expected. Plan: **weekly hCG until negative**.

**5. What would have made you choose surgery instead?**

> Significant pain, signs of rupture/haemoperitoneum, mass ≥35 mm, **heartbeat**, hCG ≥5,000, contraindications (breastfeeding, abnormal LFTs/renal function, blood dyscrasia), IUP present (heterotopic), inability to attend follow-up, or patient preference.

**Take-away:** Methotrexate is excellent for the right patient – small, quiet, low-hCG ectopic in a woman who can come back – but surgery remains the safe default when those conditions are not met.

## Case 3: 'PID' that wasn't

A 21-year-old university student attends a HC IV with 3 days of lower abdominal pain and 'some spotting'. She was treated 2 days ago at a private clinic for PID and malaria. She says her period is 'a bit late'.

**1. What are the key questions in your history?**

> **Exact LMP** and whether the 'spotting' could be mistaken for a period; site and sequence of pain (unilateral, before bleeding?); shoulder-tip pain, dizziness; fever, discharge, dysuria; previous STI/PID; contraception (e.g. implant/POP); sexual history; previous ectopic.

**2. What is the single most important bedside test and why?**

> **Urine pregnancy test** – a missed period + pelvic pain + spotting is ectopic until proven otherwise; PID and ectopic co-exist and share risk factors.

**3. hCG positive. Pulse 104, BP 108/70, left adnexal tenderness, cervical excitation. Transabdominal scan (TVUS unavailable): empty uterus, small amount of free fluid, no clear mass. What next?**

> **Suspected ectopic**. Admit; IV access; Hb, group & crossmatch; **serum hCG** (if >~6,500 with empty uterus on TAS = strongly suggestive); ask for TVUS/experienced sonographer at referral; frequent vitals. Low threshold for **laparotomy** if pain increases, Hb falls, or vitals deteriorate. If no ultrasound help, **culdocentesis** showing non-clotting blood supports haemoperitoneum.

**4. Four hours later her pulse is 122 and BP 90/60, with increasing abdominal pain. Plan?**

> Deteriorating → **ruptured ectopic**: resuscitate, crossmatch, consent, **emergency laparotomy** (salpingectomy). Do not wait for further imaging.

**5. What system lessons does this case teach?**

> Pregnancy test for all women of reproductive age with abdominal pain at every level (including private clinics/drug shops); doxycycline was also inappropriate in pregnancy; safety-net advice; referral pathways for TVUS; audit missed diagnoses.

**Take-away:** Missed period + pelvic pain = pregnancy test first; 'PID' and 'malaria' are the commonest labels under which Ugandan women with ectopics are sent home.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.